

DoubleCNN: Multivariate Time Series Prediction with Mixed Channel Information

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Abstract—Multivariate time series (MTS) forecasting is a fundamental task with applications in meteorology, traffic systems, finance, and energy management. Traditional forecasting models typically follow either channel-independent (CI) or channel-dependent (CD) strategies. CI models scale well but ignore inter-variable relations, whereas CD models capture cross-channel dependencies but often exhibit high computational overhead or unstable training dynamics.

This project investigates lightweight mixed-channel convolutional modeling and proposes a family of architectures (DoubleCNN v1–v3) that explicitly capture both temporal and cross-variable interactions using convolutional feature extractors. The proposed residual 2D mixed-channel convolution model (Double2DCNN v3) achieves strong performance on the Weather dataset, reaching a minimum MSE of 0.00584 without normalization and 0.00609 with RevIN normalization. These results demonstrate that explicit channel mixing via convolution is both effective and computationally efficient.

Furthermore, we provide comprehensive experiments comparing DoubleCNN with classical CNN, MLP, RNN, and state-of-the-art Transformer-based baselines, highlighting its efficiency and robustness.

I. Introduction

Multivariate time series (MTS) forecasting aims to predict future trajectories based on multiple correlated temporal signals. Given an input $X \in \mathbb{R}^{T \times C}$ consisting of C variables over T time steps, the goal is to estimate future values $\hat{Y} \in \mathbb{R}^{H \times C}$ over a horizon H .

Two major modeling strategies exist:

- Channel-Independent (CI) models treat each channel independently using univariate forecasting modules. They scale well but fail to capture cross-variable dependencies.
- Channel-Dependent (CD) models jointly model all variables, often using attention or graph structures. Despite modeling power, they tend to be computationally heavy and less interpretable.

Motivation. Many real-world MTS exhibit strong cross-channel correlations (temperature–humidity–pressure, traffic flows across sensors, energy load across regions, etc.). Efficiently modeling such relations without relying on attention remains a key challenge.

This work proposes DoubleCNN, a family of mixed-channel convolutional architectures that:

- 1) capture temporal patterns and cross-variable interactions simultaneously,
- 2) remain lightweight and computationally efficient,
- 3) provide interpretable features via separate temporal and channel convolutions.

Contributions

- We design a progressive family of mixed-channel convolutional architectures (DoubleCNN v1–v3), enabling explicit temporal and cross-channel feature learning.
- We introduce Double2DCNN, combining temporal and channel convolutions in residual 2D convolutional blocks.
- We provide a unified analysis of mixed-channel modeling and show its efficiency compared to classical CNN/MLP/RNN and modern Transformer baselines.
- Extensive experiments on the Weather dataset and additional benchmarks validate that mixed-channel convolution significantly improves forecasting accuracy while maintaining lightweight computation.

II. Related Work

A. Classical Statistical Models

ARIMA and VAR [1] model temporal autocorrelation but struggle with nonlinear or high-dimensional series. State-space models and exponential smoothing have similar limitations.

B. Deep Learning for Time Series

RNN/LSTM/GRU [2], [3] capture sequential dependencies but often suffer from gradient issues in long sequences. Temporal Convolutional Networks (TCN) [4] use causal convolutions to model long-range dependencies efficiently. Recent deep learning models for MTS include:

- CNN-based models: capture local temporal patterns with low computation [5], [15].
- Transformers: Informer [6], Autoformer [7], FEDformer [8], PatchTST [9], iTransformer [10] achieve

state-of-the-art results but require high computational resources.

- Other architectures: N-BEATS [16], SCINet [14], DLinear [15] demonstrate lightweight alternatives for long-sequence forecasting.

C. Channel-Mixing Models

TimeMixer [13] and Crossformer [11] explicitly model cross-channel relations via attention. Our method focuses on a lightweight alternative using mixed-channel convolutions, reducing complexity while maintaining strong cross-channel representation.

III. Methodology

Let $X \in \mathbb{R}^{C \times T}$ be the input multivariate series.

A. DoubleCNN v1: Separate Temporal and Channel Convolutions

v1 applies:

$$F_t = \text{Conv1D}_t(X), \quad F_c = \text{Conv1D}_c(X^\top),$$

where F_t captures temporal features and F_c captures inter-channel dependencies. The final representation concatenates both:

$$F = [F_t; F_c].$$

B. DoubleCNN v2: Same Padding, RevIN, and MLP Head

v2 improves stability and generalization with:

- Same-padding to maintain sequence lengths across layers.
- RevIN normalization [12] to handle distribution shifts.
- MLP prediction head for enhanced nonlinear modeling.

C. Double2DCNN v3: Mixed-Channel 2D Convolution

v3 treats X as a 2D signal and applies:

$$F_t = \text{Conv2D}(X; k_t \times 1), \quad F_c = \text{Conv2D}(X; 1 \times k_c),$$

with k_t and k_c as temporal and channel kernel sizes, respectively.

1) Residual Fusion:

$$\hat{X} = X + F_t + F_c$$

maintains the original information while adding learned temporal and channel features.

2) Weighting Fusion:

$$\hat{X} = X \odot (\sigma(F_t) \cdot \sigma(F_c))$$

emphasizes informative features but is slightly less stable empirically.

IV. Experiments

A. Experimental Setup

Datasets: Weather (main), ETTh1/ETTh2/ETTm1/ETTm2 for additional validation. Prediction length: $H = 96$.

Training: Adam optimizer, learning rate 0.001, early stopping based on validation loss, batch size 32. Experiments run on NVIDIA RTX A5000 GPU.

Evaluation Metrics:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad \text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\text{MSE}},$$

B. Key Experimental Findings (v1–v3)

1) v1: Separate Convolutions:

- Captures temporal and channel patterns but unstable.
- Best MSE ≈ 0.0115 on Weather-96.

2) v2: RevIN + MLP:

- RevIN stabilizes training.
- Best MSE reduces to **0.0038**.

3) v3: Double2DCNN (Proposed): Residual 2D mixed-channel convolution achieves:

$$\text{MSE} = 0.00584 \quad (\text{no RevIN}), \quad \text{MSE} = 0.00609 \quad (\text{with RevIN})$$

Advantages:

- lightweight,
- stable,
- robust to normalization,
- interpretable (separate temporal + channel dependencies).

C. Comparison with Baselines

TABLE I: Baseline Models on Weather (pred_len=96)

Model	MSE	MAE	RMSE	MAPE (%)
CNN	0.2875	0.3587	0.5362	12.3
MLP	0.2872	0.3550	0.5361	12.2
RNN	0.2719	0.3372	0.5214	11.7
Double2DCNN v3 (ours)	0.00584	0.0571	0.0765	1.9

Analysis: Double2DCNN reduces both MSE and MAE drastically, showing that mixed-channel convolution captures both temporal and cross-variable patterns efficiently. Compared to attention-based methods, it requires fewer parameters and lower computational cost.

D. Ablation Study

We perform ablations on:

- Fusion strategy: Residual fusion vs weighting fusion. Residual performs better and is more stable.
- RevIN usage: v3 shows robustness to normalization, though earlier versions benefit from RevIN.
- Kernel sizes: Larger temporal kernels improve long-range forecasting, larger channel kernels better capture cross-variable interactions.

V. Discussion

Key insights:

- Mixed-channel convolutions are effective for capturing cross-variable dependencies with low computational overhead.
- Residual fusion preserves original signal while integrating temporal and channel features.
- Robustness to normalization simplifies deployment in real-world datasets with varying scales.
- Compared with Transformers, our method is highly efficient for mid-length prediction horizons.
- Limitations include potential difficulty for extremely long-sequence forecasting without additional hierarchical design.

VI. Conclusion

This work systematically explores efficient channel-dependent modeling for MTS forecasting through progressive mixed-channel convolution architectures. The final Double2DCNN v3 achieves strong empirical performance while maintaining computational efficiency.

Future directions:

- Hybrid convolution-attention models for long sequences.
- Hierarchical or multi-scale mixed-channel modeling.
- Deployment on real-world systems such as energy grids, traffic networks, and financial forecasting.

References

- [1] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Wiley, 2015.
- [2] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” *NeurIPS*, 1997.
- [3] K. Cho et al., “Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation,” *EMNLP*, 2014.
- [4] S. Bai, J. Kolter, and V. Koltun, “An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling,” *arXiv:1803.01271*, 2018.
- [5] A. Borovykh, S. Bohte, and C. W. Oosterlee, “Conditional Time Series Forecasting with Convolutional Neural Networks,” in *NIPS Workshop*, 2017.
- [6] H. Zhou et al., “Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting,” *AAAI*, 2021.
- [7] H. Wu et al., “Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting,” *NeurIPS*, 2021.
- [8] H. Zhou et al., “FEDformer: Frequency Enhanced Decomposed Transformer for Long-Term Series Forecasting,” *ICML*, 2022.
- [9] X. Nie et al., “PatchTST: Time Series Patch Transformers,” *ICLR*, 2023.
- [10] Y. Liu et al., “iTransformer: Improved Transformer for Multi-variate Time-Series Forecasting,” *ICLR*, 2024.
- [11] Z. Zhang et al., “Crossformer: Efficient Cross-Variable Attention for Time Series,” *ICLR*, 2023.
- [12] H. Kim et al., “RevIN: Reversible Instance Normalization for Accurate Time-Series Forecasting,” *ICLR*, 2022.
- [13] Y. Liu et al., “TimeMixer: Lightweight Temporal and Cross-Channel Mixing for Time Series,” *ICLR 2024 Spotlight*.
- [14] Y. Liu et al., “SCINet: A Fast and Accurate Deep Learning Model for Time Series Forecasting,” *AAAI*, 2022.
- [15] H. Zeng et al., “DLinear: Towards More Efficient Long-Sequence Forecasting,” *ICLR* 2023.

- [16] B. Oreshkin et al., “N-BEATS: Neural Basis Expansion Analysis for Time Series Forecasting,” *ICLR*, 2019.

Appendix A Full Experimental Results

This appendix presents the full experimental results of DoubleCNN variants on Weather and ETTh/ETTm datasets. Results include MSE and MAE for different kernel size combinations and model settings. DTW metrics were not calculated.

A. Weather Dataset v1.0 (Baseline Convolution)

TABLE II: MSE for different kernel size combinations (v1.0, Weather dataset, pred_len=96)

t_kernel	v_kernel	1.0	7.0	14.0	21.0
1.0		0.0176	0.0187	0.0169	0.0152
16.0		0.0208	0.0239	0.0283	0.0195
32.0		0.0508	0.0369	0.0348	0.0279
64.0		0.0220	0.0188	0.0261	0.0232
96.0		0.0115	0.0123	0.0141	0.0144

TABLE III: MAE for different kernel size combinations (v1.0, Weather dataset, pred_len=96)

t_kernel	v_kernel	1.0	7.0	14.0	21.0
1.0		0.1038	0.1072	0.1014	0.0970
16.0		0.1120	0.1206	0.1327	0.1090
32.0		0.1747	0.1489	0.1474	0.1276
64.0		0.1124	0.1045	0.1245	0.1092
96.0		0.0837	0.0855	0.0911	0.0917

B. v1.1: Adding RevIN Module

TABLE IV: MSE for different kernel size combinations (v1.1, Weather dataset, pred_len=96)

t_kernel	v_kernel	1.0	7.0	14.0	21.0
1.0		0.0227	0.0218	0.0189	0.0184
16.0		0.0240	0.0259	0.0253	0.0240
32.0		0.0394	0.0353	0.0465	0.0240
64.0		0.0238	0.0286	0.0252	0.0166
96.0		0.0146	0.0150	0.0149	0.0133

TABLE V: MAE for different kernel size combinations (v1.1, Weather dataset, pred_len=96)

t_kernel	v_kernel	1.0	7.0	14.0	21.0
1.0		0.1174	0.1155	0.1078	0.1068
16.0		0.1194	0.1250	0.1236	0.1209
32.0		0.1512	0.1434	0.1640	0.1149
64.0		0.1197	0.1266	0.1207	0.0987
96.0		0.0943	0.0952	0.0951	0.0902

C. v2.0: Same-padding + MLP Head

TABLE VI: MSE for different kernel size combinations (v2.0, Weather dataset, pred_len=96)

t_kernel	v_kernel	1.0	7.0	15.0	21.0
1.0		0.0149	0.0074	0.0102	0.0081
15.0		0.0083	0.0145	0.0069	0.0090
31.0		0.0241	0.0087	0.0113	0.0181
63.0		0.0088	0.0078	0.0073	0.0075
95.0		0.0130	0.0229	0.0285	0.0148

TABLE VII: MAE for different kernel size combinations (v2.0, Weather dataset, pred_len=96)

t_kernel	v_kernel	1.0	7.0	15.0	21.0
1.0		0.0865	0.0676	0.0803	0.0690
15.0		0.0724	0.0974	0.0651	0.0732
31.0		0.1270	0.0731	0.0823	0.1019
63.0		0.0734	0.0688	0.0677	0.0673
95.0		0.0897	0.1107	0.1034	0.0930

D. v2.1: GELU Activation

TABLE VIII: MSE for different kernel size combinations (v2.1, Weather dataset, pred_len=96)

t_kernel	v_kernel	1.0	7.0	15.0	21.0
1.0		0.0154	0.0074	0.0109	0.0053
15.0		0.0071	0.0076	0.0055	0.0108
31.0		0.0099	0.0051	0.0085	0.0038
63.0		0.0062	0.0052	0.0071	0.0073
95.0		0.0115	0.0129	0.0109	0.0110

TABLE IX: MAE for different kernel size combinations (v2.1, Weather dataset, pred_len=96)

t_kernel	v_kernel	1.0	7.0	15.0	21.0
1.0		0.0967	0.0674	0.0782	0.0569
15.0		0.0649	0.0687	0.0585	0.0699
31.0		0.0792	0.0560	0.0707	0.0484
63.0		0.0626	0.0571	0.0637	0.0646
95.0		0.0829	0.0862	0.0782	0.0831

E. v3.0: DoubleCNN v3.0

TABLE X: DoubleCNN v3.0 performance on Weather dataset (pred_len=96)

Model Variant	MSE	MAE
Residual, norm	0.00609697	0.0617271
Weighted, norm	0.0223475	0.1258370
Residual, no norm	0.00583928	0.0602142
Weighted, no norm	0.0148058	0.0978114

F. ETTh1, ETTh2, ETTm1, ETTm2 Datasets

TABLE XI: DoubleCNN v3.0 performance on ETTh1 dataset (pred_len=96)

Model Variant	MSE	MAE
Residual, norm	1.568615	1.204032
Weighted, norm	1.720398	1.265438
Residual, no norm	1.568615	1.204032
Weighted, no norm	1.720398	1.265438

TABLE XII: DoubleCNN v3.0 performance on ETTh2 dataset (pred_len=96)

Model Variant	MSE	MAE
Residual, norm	1.424018	1.040097
Weighted, norm	1.387440	1.037303
Residual, no norm	1.424018	1.040097
Weighted, no norm	1.387440	1.037303

TABLE XIII: DoubleCNN v3.0 performance on ETTm1 dataset (pred_len=96)

Model Variant	MSE	MAE
Residual, norm	1.531612	1.188507
Weighted, norm	1.588784	1.217913
Residual, no norm	1.531612	1.188507
Weighted, no norm	1.588784	1.217913

TABLE XIV: DoubleCNN v3.0 performance on ETTm2 dataset (pred_len=96)

Model Variant	MSE	MAE
Residual, norm	1.318804	0.996933
Weighted, norm	1.375558	1.032235
Residual, no norm	1.318804	0.996933
Weighted, no norm	1.375559	1.032235

G. Baseline Models on Weather Dataset

TABLE XV: Baseline models performance (Weather dataset, pred_len=96)

Model	MSE	MAE
CNN	0.287492	0.358667
MLP	0.287247	0.355004
RNN	0.271915	0.337192

Note: DTW metrics were not calculated.